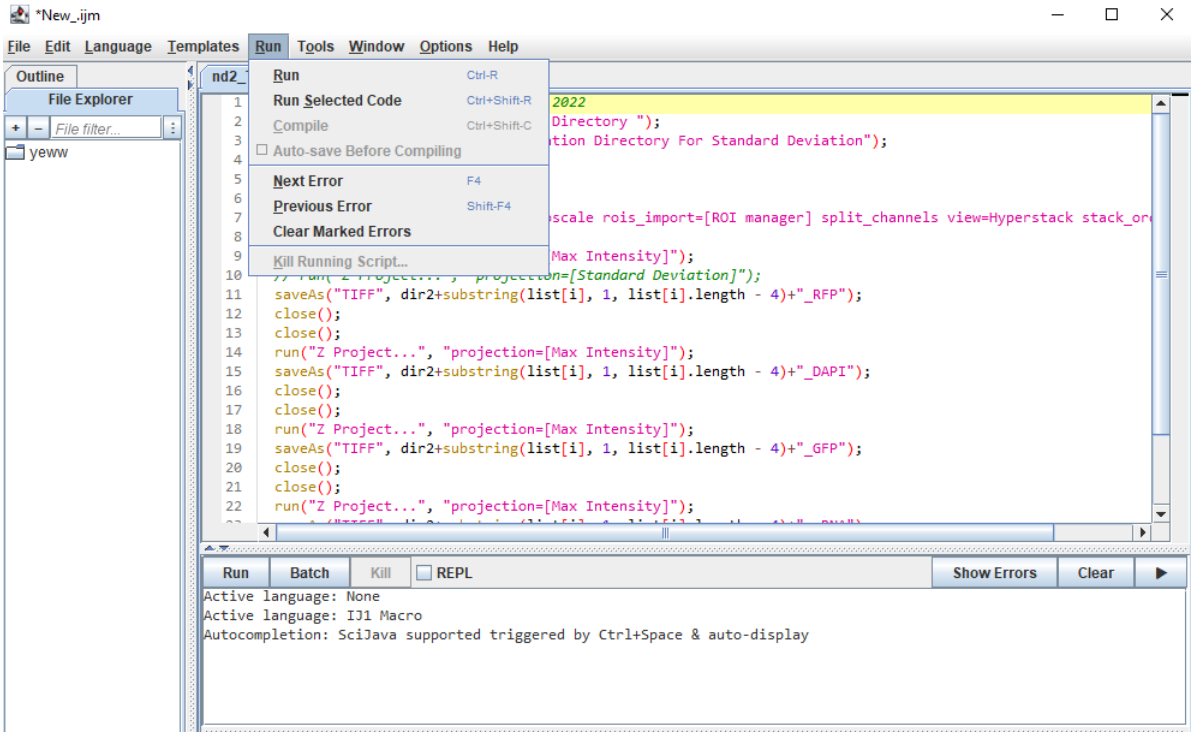
1 Getting the necessary software

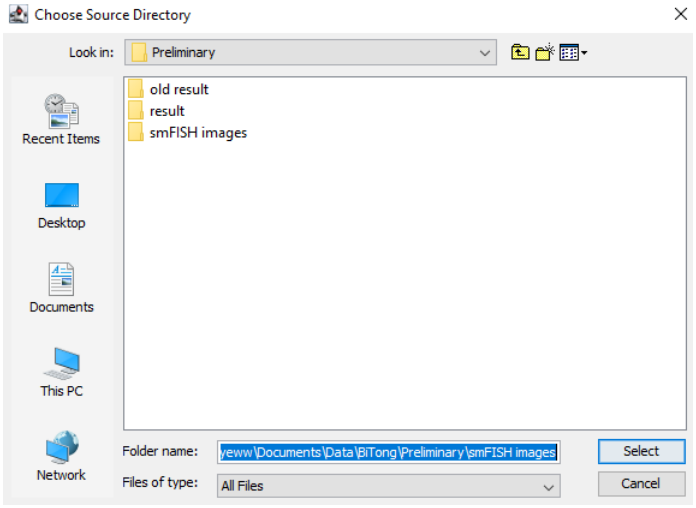
1. Download Fiji's distribution of the ImageJ from
   `https://imagej.net/software/fiji/downloads`.
2. Download CellProfiler from
   `https://cellprofiler.org/releases`.
3. Download MATLAB from
   `https://au.mathworks.com/login?uri=%2Fdownloads%2Fweb_downloads`.

2 Extracting the channels

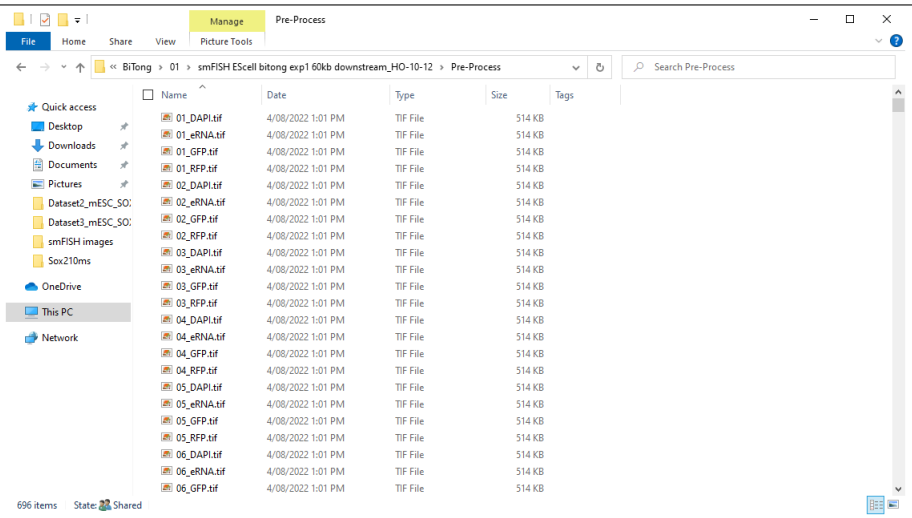
4. Start up Fiji.
5. Go File - Open.
6. Select the *nd2_To_tif.ijm* file.
7. A new window like the one shown below should appear, go Run - Run.



8. Another new windows as follow will pop up, select the folder that contains the .nd2 files (make sure the folder only contains the nd2 files of interest and nothing else).



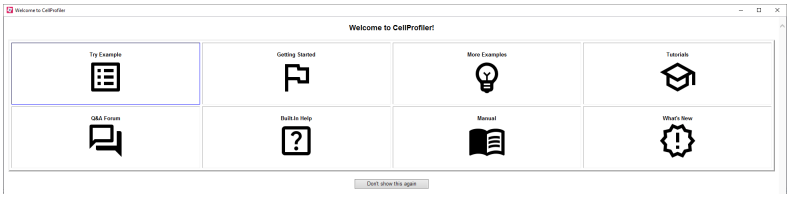
9. After that, another similar window will pop up. Select a folder to store the new images.
10. If everything is done correctly, your new folder should fill with images of different channels as follows.



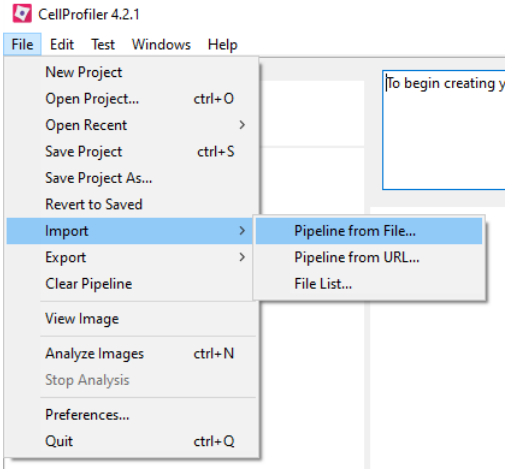
11. You may now exit Fiji.

3 Identify the objects in each channel

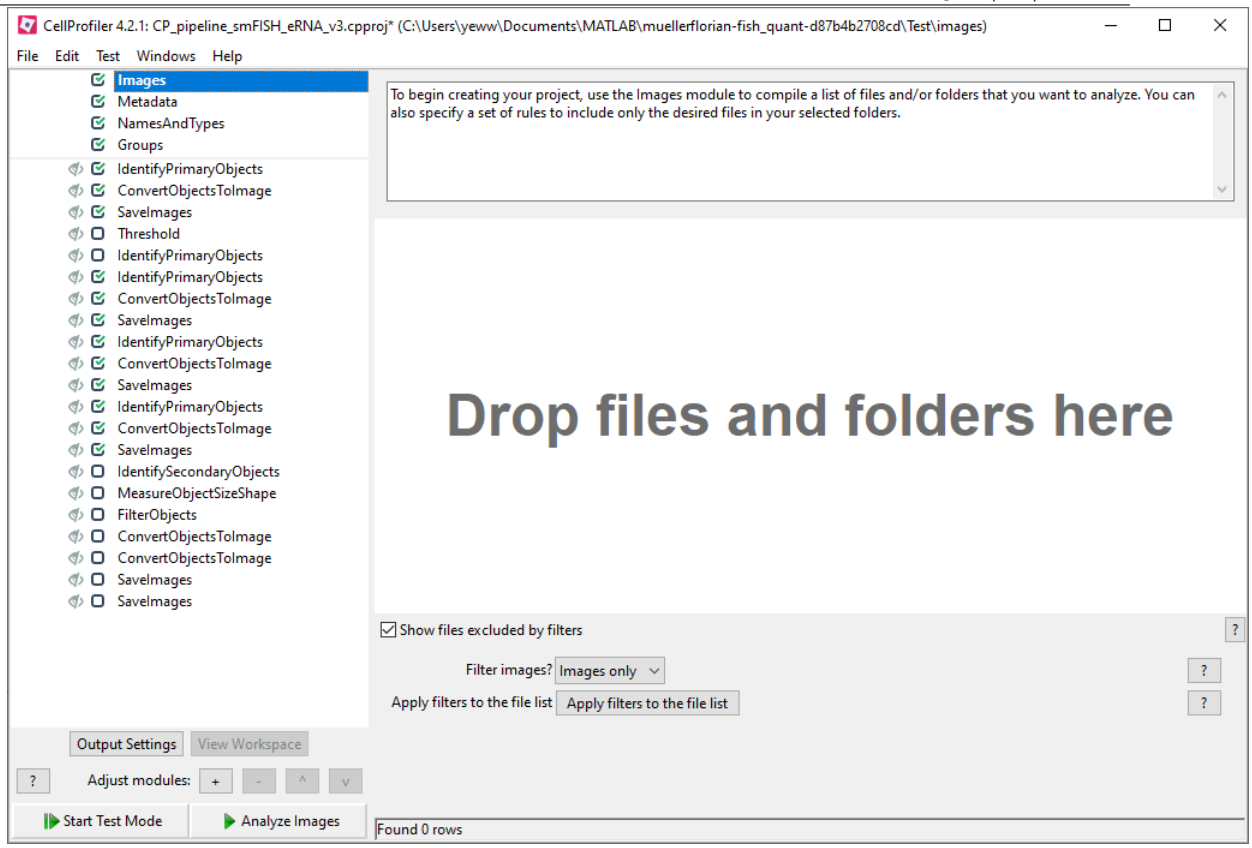
12. Run CellProfiler.
13. If a "Welcome to CellProfiler" windows pop up (such as shown below), feels free to close it.



14. Go File - Import - Pipeline from File... as shown in the figure below.

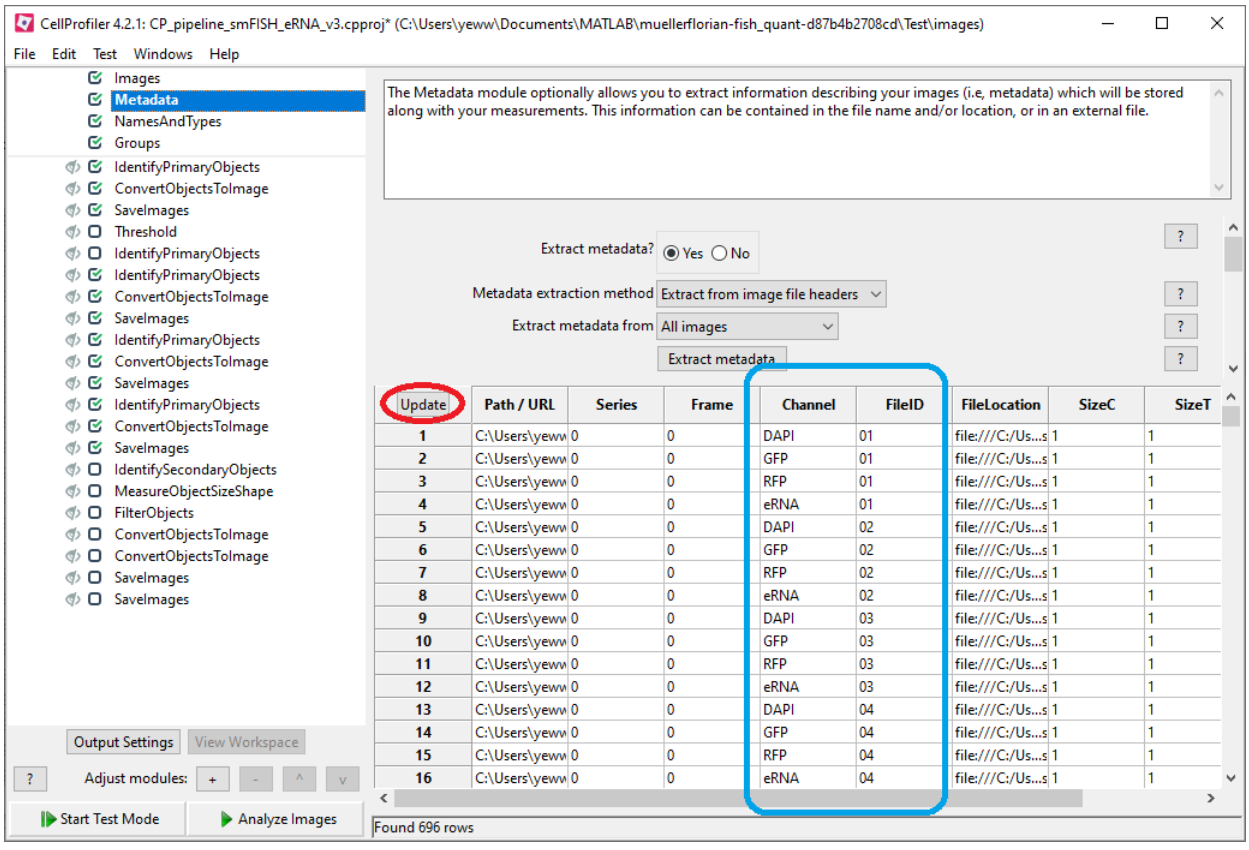


15. Select the *CP_pipeline_smFISH_eRNA_v3.cppipe* file.
16. Your CellProfiler should now update to be similar to the figure below. Click on the "Image" on the left hand side (the one highlighted in the image below).

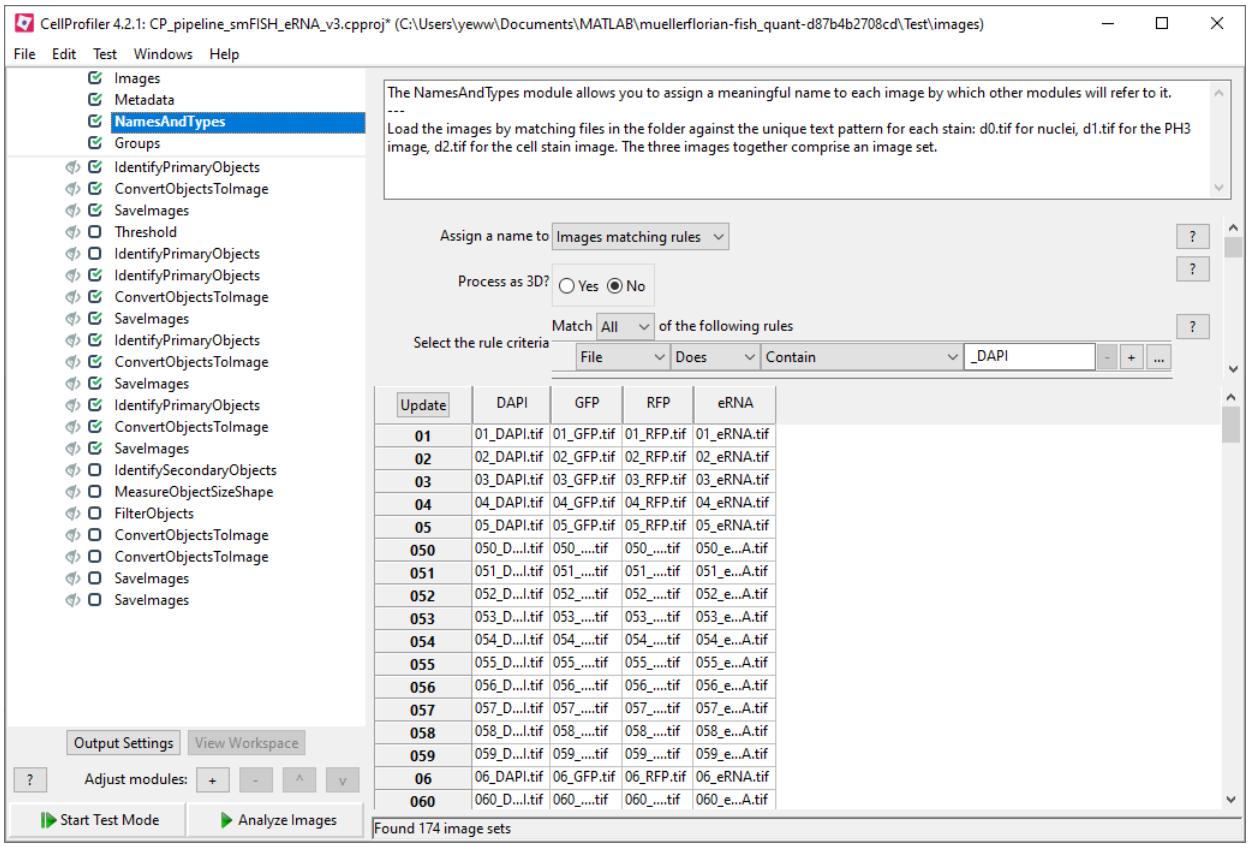


17. Drag all the images produced in Step 10 to the "Drag files and folders here".

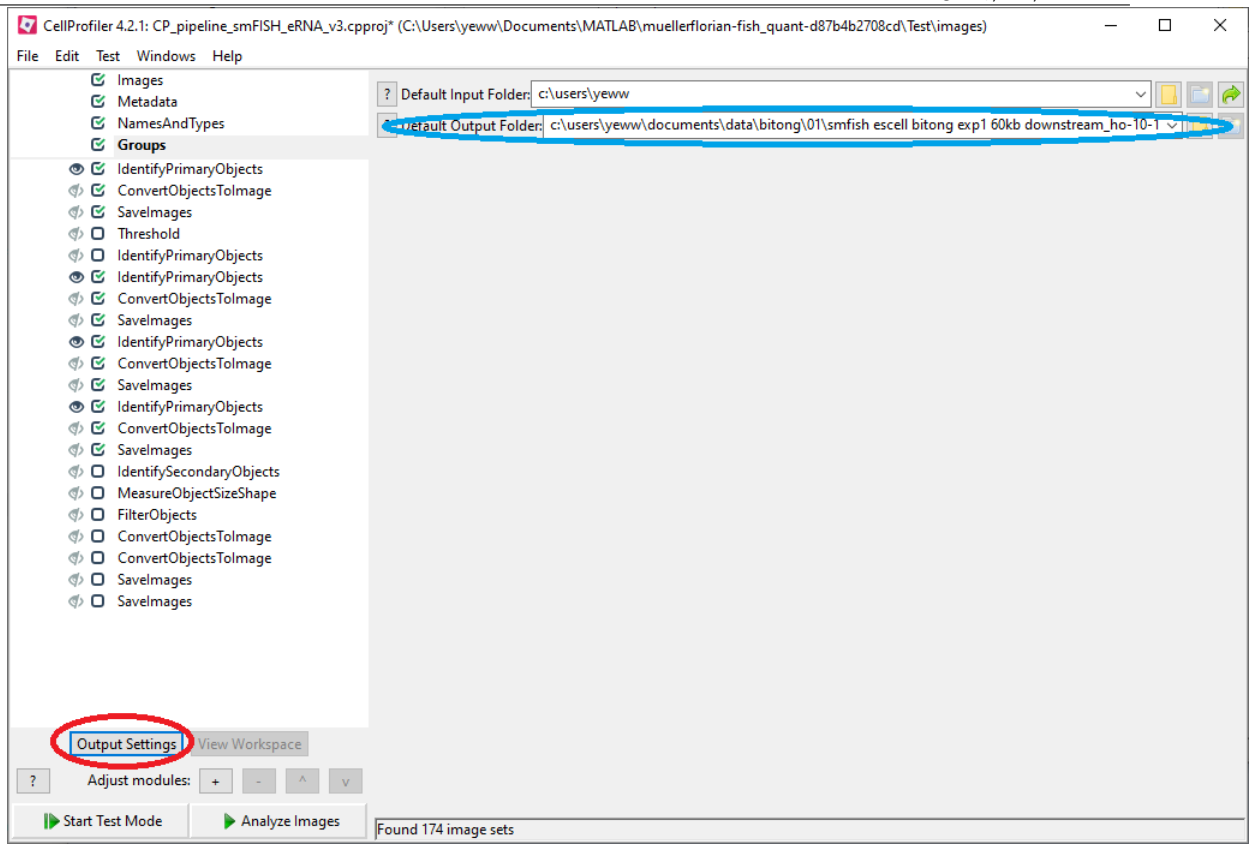
18. Click on the "Metadata" on the left hand side and click the "Update" as shown in the red circle. Ensure that the "Channel" and "FileID" has been correctly identified as shown in the blue area.



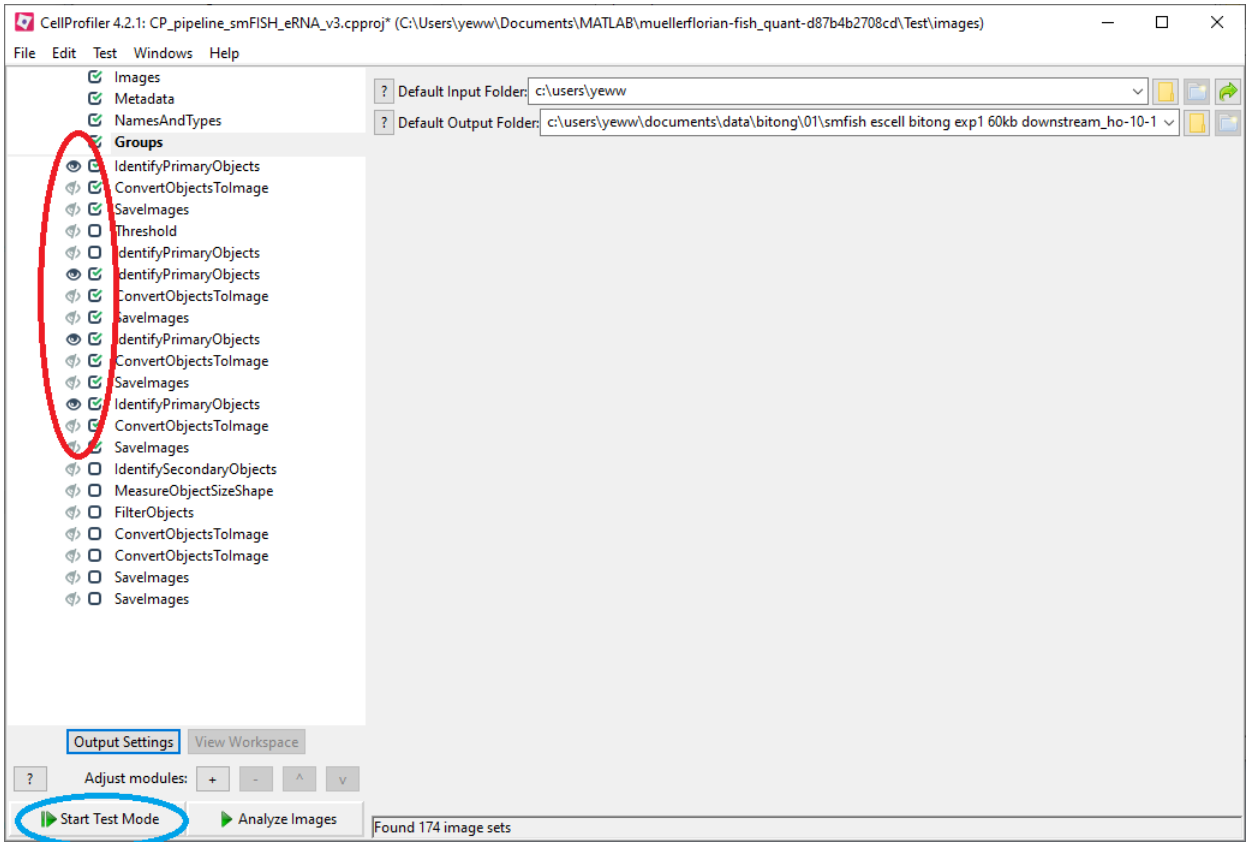
19. Click on the "NamesAndTypes" on the left hand side and click the "Update" as shown in the red circle. Ensure that the files are correctly identified as shown in the image below.



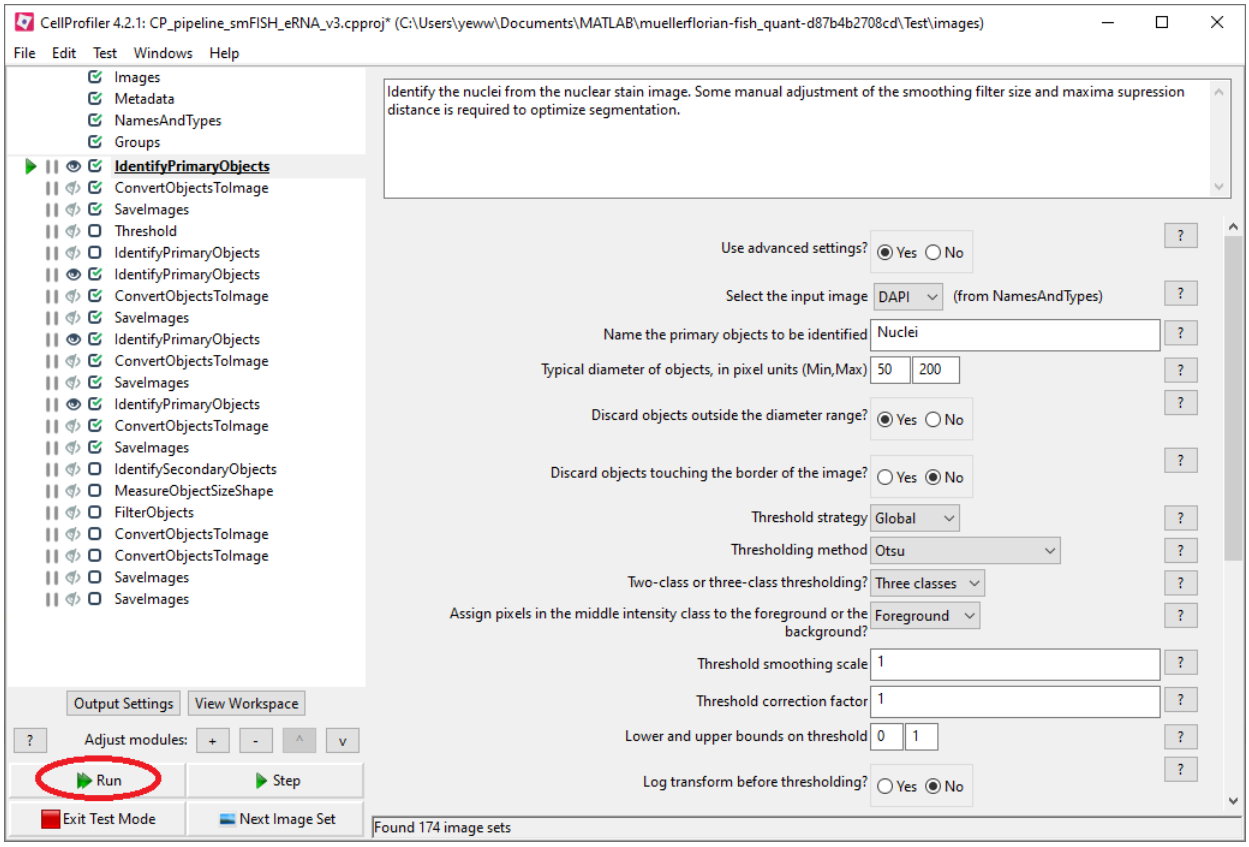
20. Click on the "Output Settings" on the bottom left hand side (marked by red circle in the image below). Ensure the "Default Output Folder:" is set to an appropriate folder you wanted.



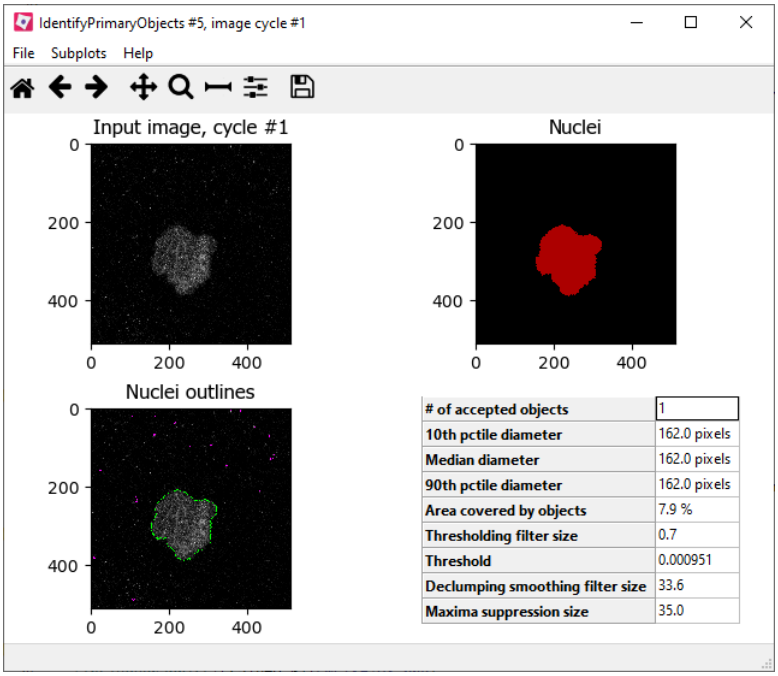
21. Click to highlight the eyes next to the "IdentifyPrimaryObjects" that have green ticks next to them to achieve something as shown by the image (highlighted by red) below. Click on "Start Test Mode" after.



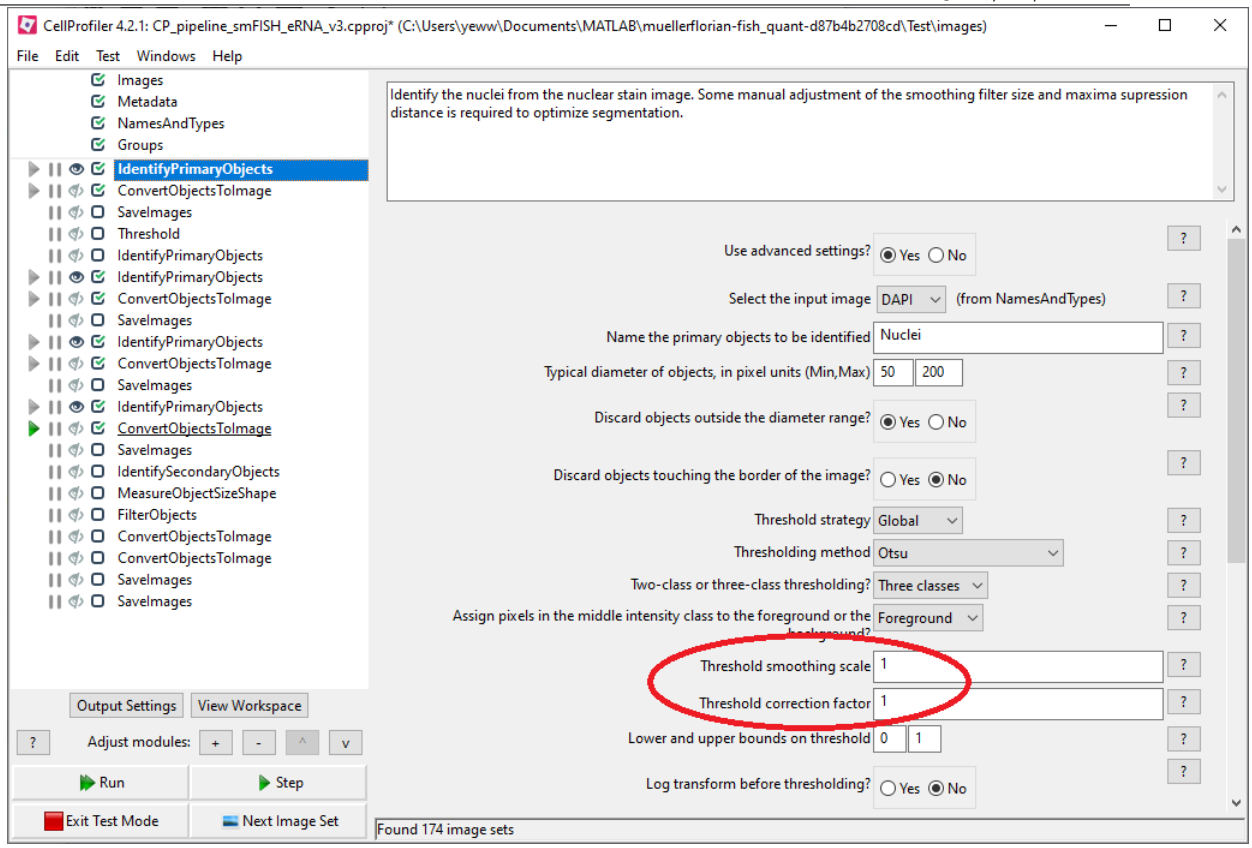
22. Click on "Run" and 4 new windows will pop up.



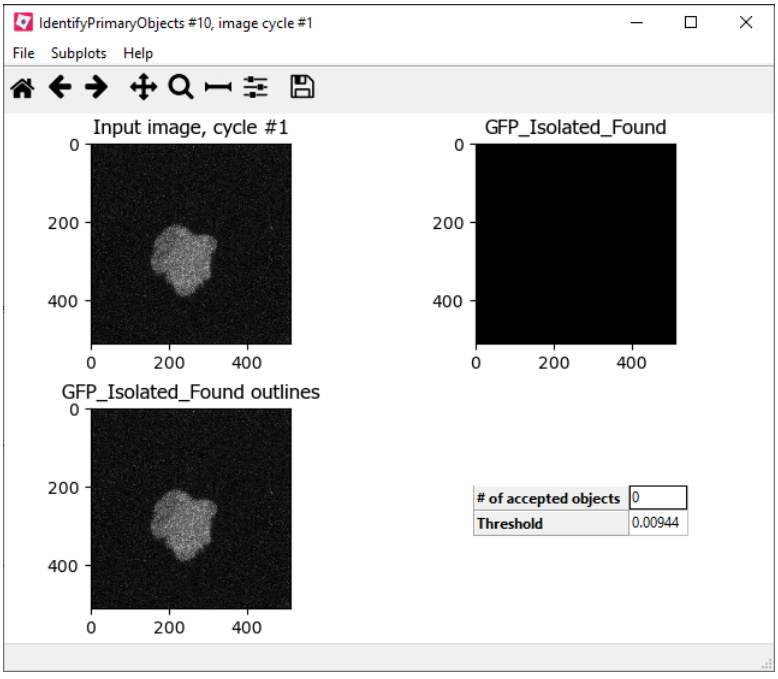
23. The first image is in respond to the "DAPI" channel as shown below.



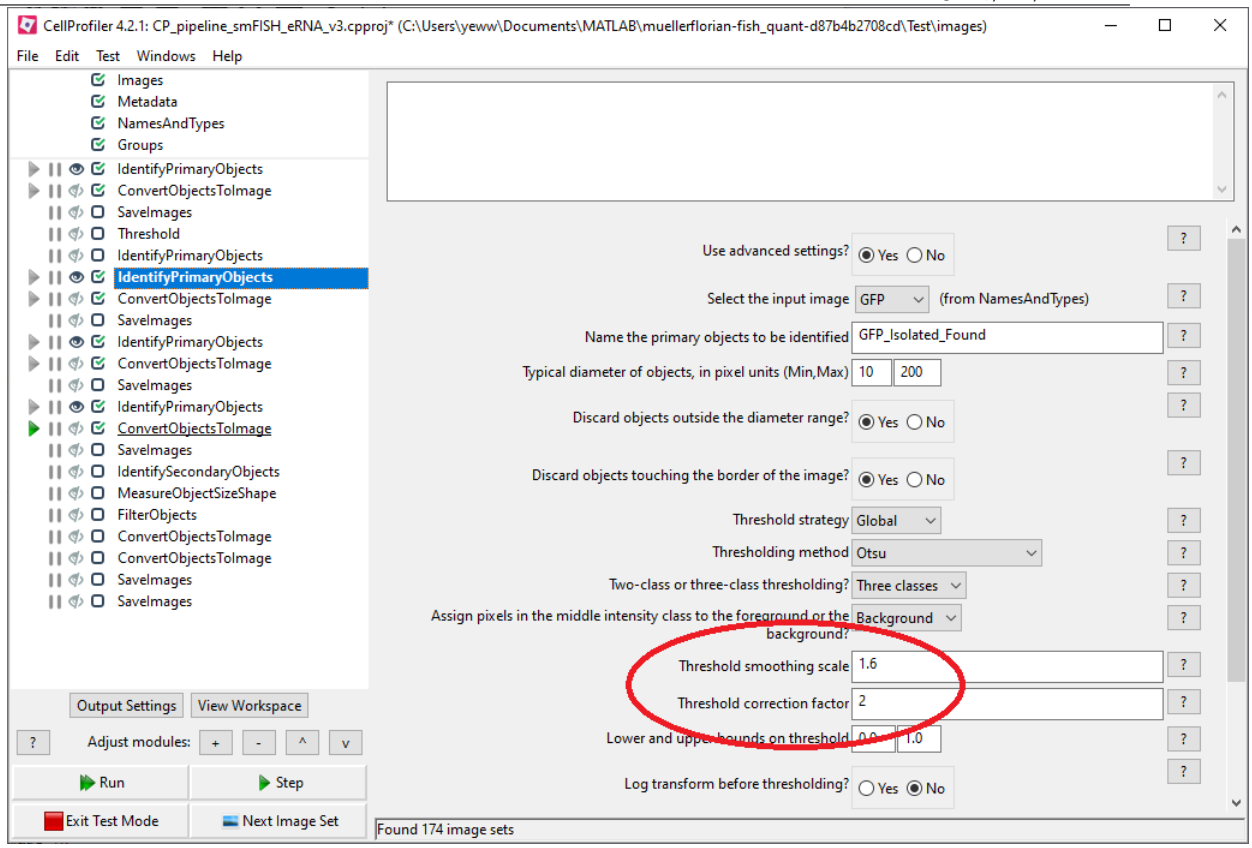
24. If the image did not correctly identify the DAPI in the bottom left and top right of the image in the new window, click the first "IdentifyPrimaryObjects" as highlighted in the image below and edit the "Threshold smoothing scale" and "Threshold correction factor" and click "Run" on the bottom right to see if it adjusted to better. Move on to second image when you're happy with the result.



25. The second image is in respond to the "GFP" channel as shown below.

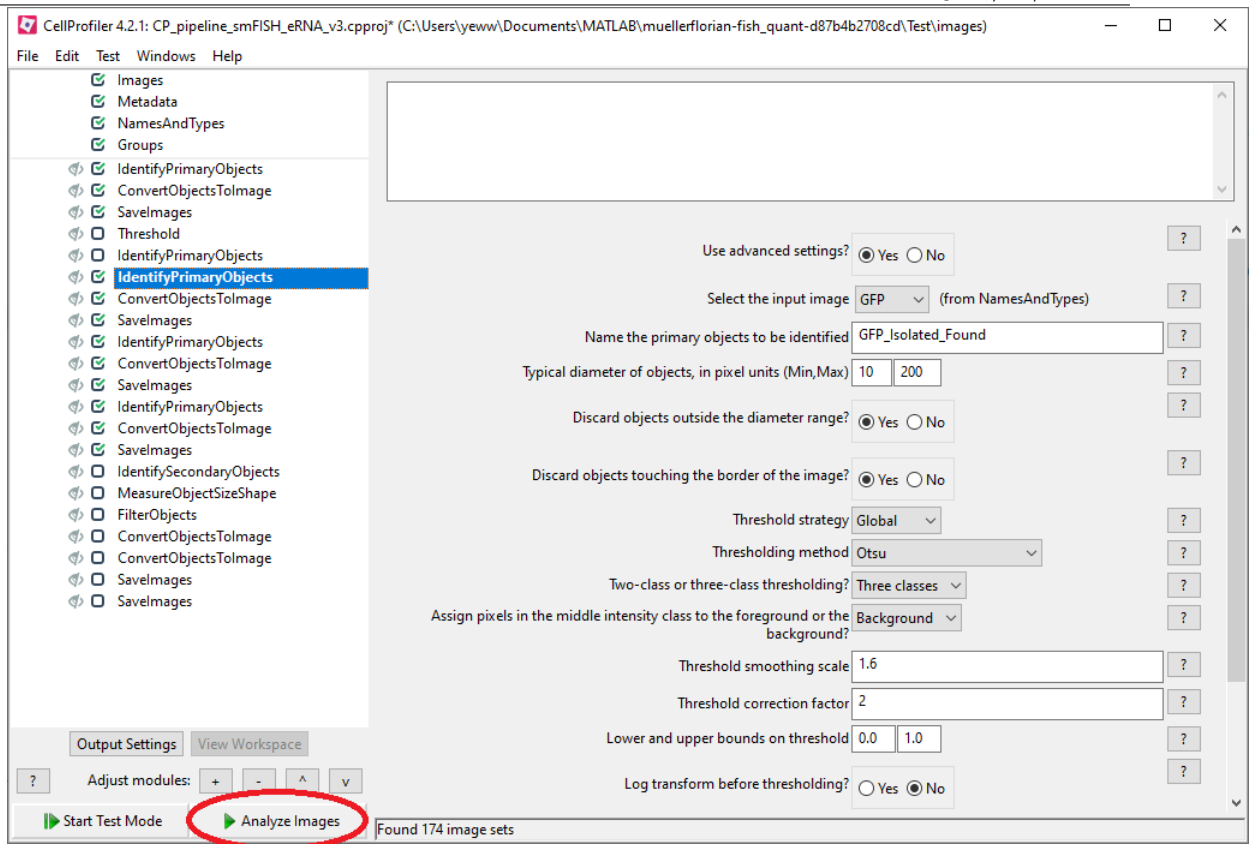


26. Similar to before, if the image did not correctly identify the GFP in the bottom left and top right of the image in the new window, click the second "IdentifyPrimaryObjects" as highlighted in the image below and edit the "Threshold smoothing scale" and "Threshold correction factor" and click "Run" on the bottom right to see if it adjusted to better. Move on to second image when you're happy with the result.

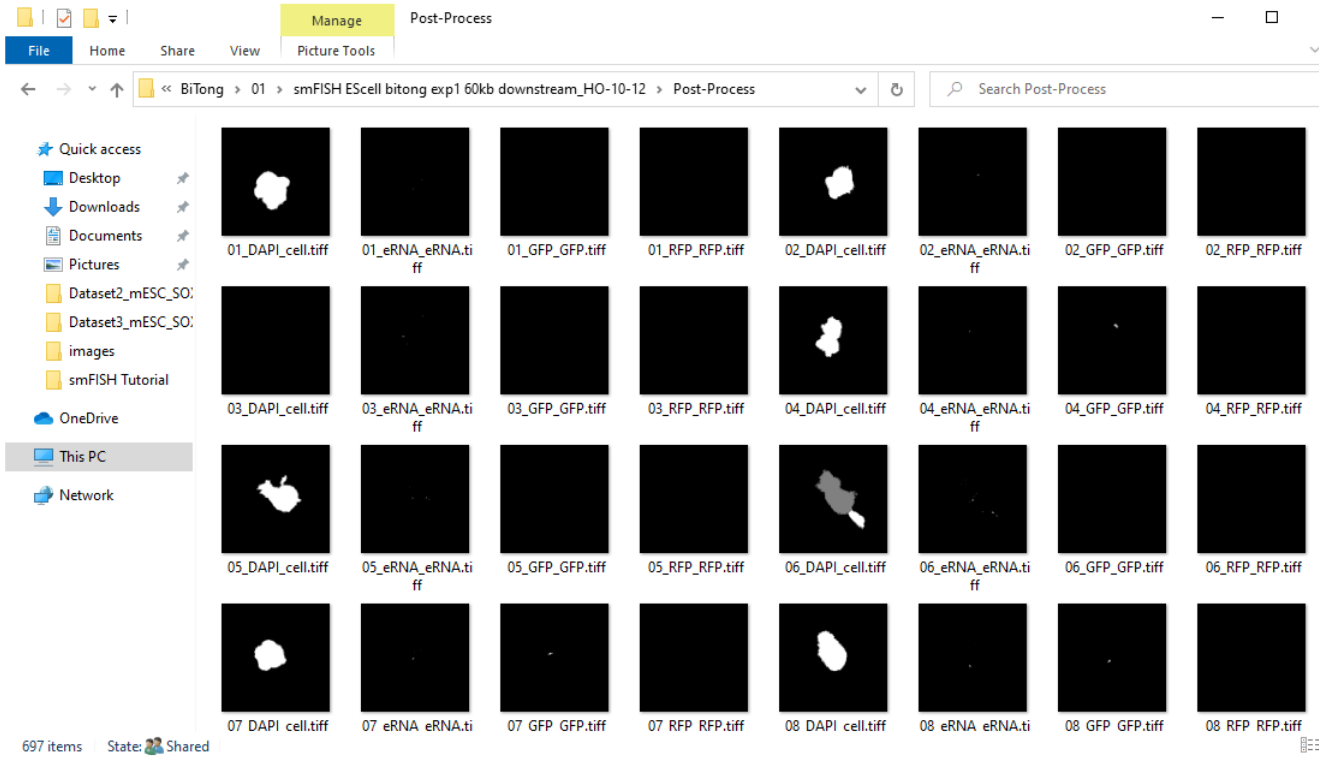


27. Repeat for the third and fourth windows. Once you're happy with this image set, click the "Next Image Set" and repeat Step 22-26. Note that if you change the value, it will affect the first image set, the target here is to capture the best behaviour across all the image sets with one setup.

28. Once you're happy with everything, click "Exit Test Mode" on bottom left, turn off the eyes and click "Analyse Images".



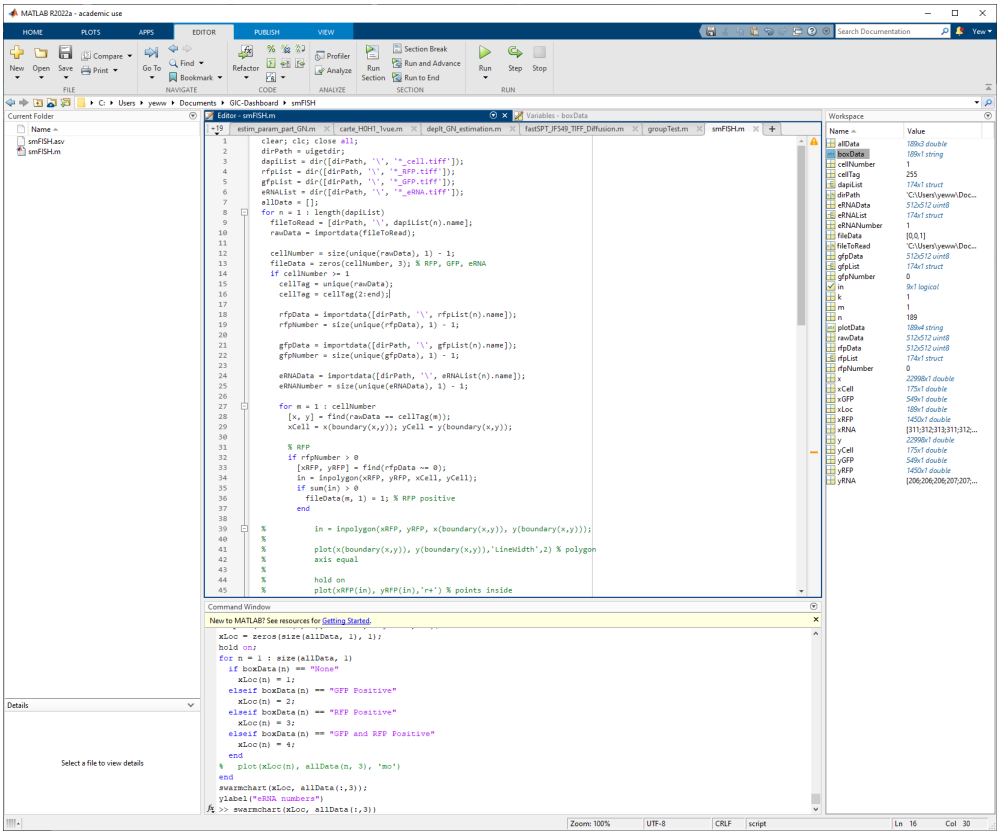
29. If you've done the steps correctly, the designated folder will contain the masks of the objects detected as shown below.



4 Cell Quantification

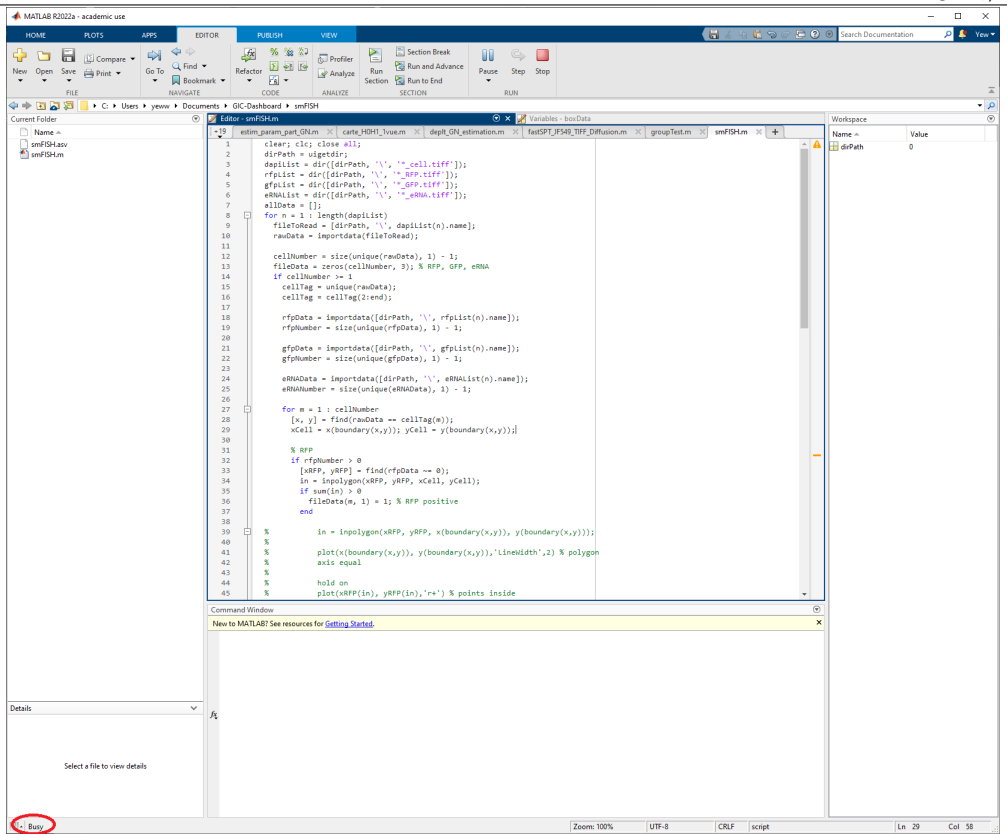
30. Start up MATLAB by running the *smFISH.m* file.

31. If you installed MATLAB correctly, you'll see a window as shown below.



32. Press "Run" and a new window will pop up asking for the location of the mask images. Select the appropriate folder.

33. Check the bottom left of the MATLAB window, it should now have a "Busy" word as shown in the image below. Let it do its things.



34. Once it's done. A new window like the one below will pop up. Feels free to "File - Save As" to store the image.

